

Niagara N4 API Comparison

oBIX | Gline REST API | BTIB ACTIVE-API | Tridium REST | Third-Party

Side-by-side comparison of five approaches to interact with Niagara N4 stations

Based on station 192.168.2.97 (openClawTestRestful) and 192.168.2.112 (IOT_Test)

Generated: 2026-06-10

Vendor: Shanghai Gline Net Co.

DISCLAIMER: This document is AI-generated based on publicly available internet research, training data, and hands-on testing on specific station versions. Features, availability, and pricing may vary by station version (Niagarax vs standard N4), configuration, license tier, and time of inquiry. Always verify directly with each vendor for your specific requirements. The AI author assumes no responsibility for inaccuracies or omissions.

Feature Comparison Table

Color legend: **Green = Yes / Full support**, **Red = No / Not supported**

Feature	oBIX (free, built-in)	Gline REST (free, custom)	BTIB ACTIVE-API (\$462~\$1k)	Tridium /rest/ (needs SMA)	Analytics (OSS/J2)
Auth method	SCRAM-SHA-256 or HTTP Basic	None (LAN only)	HTTP Basic / Token	HTTP Basic	API Key / OAuth2
Read values	GET /obix/config/...	GET /api/config?path=	GET /api/v1/...	GET /rest/...	Yes (read-only)
Write values	PUT to point	POST /api/batch (MOD)	PUT .../value	PUT .../{slot}	No
Create components	No	Yes (ADD) any type	Yes POST components	Limited	No
Delete components	No	Yes (REMOVE)	Yes DELETE	No	No
Deep Copy	No	Yes (CP) rebuilds links	No	No	No
Rename (link-safe)	No	Yes (RENAME) handle ORD	No	No	No
Link / Unlink	POST create	Yes (LINK/UNLINK)	Yes	No	No
Invoke actions	POST to action	Yes (INVOKE)	Yes (POST)	No	No
Batch operations	No	Yes (10 ops) atomic POST	Limited (multi-query)	No	No
Alarm query	POST /~alarmQuery/	No	Yes dedicated API	No	Limited
History query	GET /~historyQuery/	No	Yes history API	Yes	Yes
License cost	Free (built-in)	Free (custom module)	\$462~\$1,001	SMA subscription	SaaS subscription
SMA required	No	No	Yes	Yes	Yes
Port	80 / 443	8081 (config.)	Configurable	80 / 443	HTTPS
Protocol	XML / JSON	JSON only	JSON	JSON	JSON

API Details & Capabilities

1. oBIX (Open Building Information Exchange)

- **Type:** Standard protocol (OASIS) - built into every Niagara N4 station
- **Best for:** Reading point values, writing setpoints, alarm queries, history data
- **Auth:** SCRAM-SHA-256 (5-step handshake) or legacy HTTP Basic (192.168.2.97)
- **Limitations:** Cannot create/delete/rename/copy components. No batch operations. One request per operation.
- **Port:** 80/443 (shares station web server port)
- **Complements Gline REST:** Use oBIX for alarm+history, Gline REST for component lifecycle.

2. Gline REST API (niagaraRestApi)

- **Type:** Custom Niagara N4 module - lightweight JDK HttpServer embedded in station
- **Best for:** Full station automation - batch create/delete/copy/rename/link components
- **Endpoints:** GET /api/status, /api/modules, /api/types, /api/config; POST /api/batch
- **10 batch operations:** ADD, LINK, RENAME, CP (deep copy), MOD, UNLINK, REMOVE, READ, LIST, INVOKE
- **No SMA required:** Works on any N4.14 station without Software Maintenance Agreement
- **No auth:** Internal LAN only - add nginx reverse proxy with Basic Auth for production
- **Alarm/History:** Not built-in - pair with oBIX for complementary coverage
- **Station:** openClawTestRestful (N4.14.0.162) @ 192.168.2.97:8081

3. BTIB ACTIVE-API

- **Type:** Commercial Niagara module - comprehensive RESTful JSON API
- **Best for:** Production deployments needing full CRUD with commercial support
- **Pricing:** Light Edition \$462 (500 points), Normal Edition \$1,001 (unlimited)
- **Capabilities:** Create/delete components, read/write values, alarm management, history, Swagger docs
- **Limitations:** No atomic batch operations; no deep copy/rename; requires SMA + BTIB license
- **Resources:** niagamarketplace.com, doc.vayandata.com

4. Tridium N4 Built-in REST (/rest/)

- **Type:** Official Tridium REST interface - available on SMA-licensed stations
- **Best for:** Simple read/write operations on N4 stations with active SMA
- **Availability:** Inconsistent - present on standard N4, may be missing on Niagara builds
- **Capabilities:** Read/write point values, some component creation, history data
- **Limitations:** No delete/copy/rename; no batch; no alarm; SMA required

5. Third-Party Analytics Platforms (OSS / Jasper / J2Innovations)

- **Type:** SaaS/on-premise analytics platforms - read-only data extraction
- **Best for:** Dashboards, energy analytics, fault detection, reporting
- **Limitations:** Read-only (cannot create/modify station components); no batch operations on station side

Decision Guide

Use oBIX when:

- You only need to read/write point values and browse the station tree.
- You need alarm and history data extraction (built-in, no extra cost).
- You cannot install custom modules on the target station.
- Single-request-per-operation is acceptable.
- You are building standard BMS integration.

Use Gline REST API when:

- You need full component lifecycle management (create, delete, copy, rename, link/unlink, invoke).
- You need atomic batch operations - up to 10 operations in one request.
- The target station does not have active SMA (Gline REST works without it).
- You can keep the API on internal LAN.
- You are a developer doing station automation scripting.

Use BTIB ACTIVE-API when:

- Your budget covers the license fee (\$462~\$1,001).
- You need commercial support and guaranteed documentation.
- The station already has active SMA.
- You need dedicated alarm and history REST endpoints.

Use Tridium /rest/ when:

- Your station has SMA and the /rest/ endpoint is available.
- You only need simple JSON read/write without complex lifecycle management.

Use third-party analytics when:

- Your need is dashboards, analytics, reporting - not station management.
- You do not need to create or modify station components.

Complementary approach: Gline REST API (component lifecycle + batch) + oBIX (alarms + history) = complete coverage for free.

capabilities directly with each vendor before making purchasing or integration decisions.

For commercial customization, integration, or deployment support, please contact:

Shanghai Gline Net Co.

Email: jason.zhang@gline-net.com

Tel / WhatsApp: +86 13801909968

--- End of Document ---